

Multiobjective Optimization Problem

- Decision vector:
$$x \in \mathcal{X} \subseteq \mathbb{R}^n$$
- Objectives:
$$\min_{x \in \mathcal{X}} \mathbf{F}(x) = (f_1(x), f_2(x), \dots, f_m(x))$$
- Constraints (physical limits and conservation of mass):
$$g_i(x) \leq 0, \quad i = 1, \dots, p, \quad h_j(x) = 0, \quad j = 1, \dots, q$$
- Optimality: Pareto dominance (strictly better in at least one objective and not worse in others):

Pareto solutions are those that are not strictly dominated by any other feasible solution.

Scalarization or Sampling in the Pareto Front

- Weighted sum for convex Pareto front:
$$\Phi_{\text{WS}}(x; \mathbf{w}) = \sum_{k=1}^m w_k f_k(x), \quad w_k \geq 0, \quad \sum_{k=1}^m w_k = 1.$$
- Tchebycheff for non-convex Pareto front:
$$\Phi_{\infty}(x; \mathbf{w}, z^*) = \max_{k=1, \dots, m} w_k |f_k(x) - z_k^*|.$$
- Optimize one objective by applying Epsilon-Constraint to others:
$$\min f_1(x) \quad \text{s.t.} \quad f_k(x) \leq \varepsilon_k, \quad k = 2, \dots, m.$$

Pareto Front - Decision

- The set of non-dominated trade-off solutions:
$$\mathcal{P} = \{x \in \mathcal{X} \mid \nexists y \in \mathcal{X} : F(y) \prec F(x)\}.$$

Evolutionary Multiobjective Optimization

- Key: populations, variation operators, and dominance-based sorting to approximate the Pareto front.
- Non-dominated Sorting: Individuals are ranked by Pareto fronts: $F_1, F_2, \dots$
- Crowding Distance: A diversity-preserving metric that measures the perimeter of the rectangle around an individual in objective space. For each objective k partial distances are summed to find the crowded region:
$$\tilde{f}_k(x)$$
- Constraint handling: Penalty-based objective

Gradient-based Optimization

- Weighted Lagrangian:
$$\mathcal{L}(x, \lambda) = \sum_k w_k f_k(x) + \sum_{i=1}^p \mu_i g_i(x) + \sum_{j=1}^q \nu_j h_j(x).$$
- Karush-Kuhn-Tucker (KKT) condition:

Bayesian Optimization

- Gaussian Process prior:
$$y(x) \sim \mathcal{GP}(\mu(x), k(x, x')).$$
- Gaussian Process Posterior:
$$\hat{\mu}_n(x') = \mu(x') + k(x', X)K^{-1}(\mathbf{y} - \mu(X))$$
- Mean (the best estimates):
- Variance (model's uncertainty):
$$\sigma_n^2(x') = k(x', x') - k(x', X)K^{-1}k(X, x').$$
- Acquisition Functions
- Expected Improvement (EI):